Cache Geometry Under Obfuscation: Per-Head Rotation Does Not Remove the Signal, and a KV-Cloak Implementation Is Inconclusive

Thomas Edrington[1]

[1]Liberation Labs
Aided by AI research agents.

Version 1.1, October 2026

Abstract

The confabulation signal that singular-value features read from a model's key-value (KV) cache survives per-head orthogonal rotation, the simulation of KV-Cloak's obfuscation [Luo et al., 2026] used in this paper, on both models where it was tested. On a Claude-distilled Qwen3.5-27B (7 confabulations, 87 hedged replies), the feature that detected best without obfuscation, the spectral gap σ_1/σ_2, goes from AUROC 0.903 to 0.787 under the rotation ($p = 0.010$); on Qwen2.5-7B (48 and 52), signal fraction goes from 0.661 to 0.639 ($p = 0.017$), or 0.765 and 0.670 after removing reply length. The rotation shifts the spectrum of the layout the lab's feature extractor uses, which stacks heads along the token axis, but it cannot change the spectrum of any single head: in a 100-prompt replication, no per-head or feature-axis feature moved. Under our implementation of the full KV-Cloak mechanism, $S \cdot \hat{P} \cdot (K + A) \cdot M$, the spectral gap falls to 0.614 ($p = 0.33$). Signal rank stays at 0.766, but it mostly tracks reply length; after removing length the best of the five features is 0.705, and adjusted for choosing it, $p = 0.19$. With seven confabulations, this shows neither that detection survives the full mechanism nor that it is removed. Injection through the cloak was not tested in any interpretable way: direct steering, the positive control, corrected 0–1 of 10 replies per model.

Correction note (version 1.1, October 2026). Version 1.0, titled "Cache Geometry Under Obfuscation: KV-Cloak as Defense Against Adversarial KV-Cache Steering" (in the lab's published-research repository on GitHub since 29 April 2026; its academic edition has been on Zenodo since June 2026, all versions 10.5281/zenodo.20596922), claimed that KV-Cloak defends cache-geometry methods against adversaries. Its abstract opened: "A single invertible linear transform of the KV cache degrades SVD-based cognitive-state detection from near-ceiling to near-chance ($p = 5.5 \times 10^{-16}$), while a real KV-Cloak implementation confirms the defense is conservative (detection AUROC drops from 0.903 to 0.614)." We withdraw that sentence and the claims built on it. The p-value tests how far the rotation shifts one feature's values, not detection; under the rotation the same feature still detects (0.787); and under the full mechanism, seven confabulations cannot tell a removed signal from a surviving one. Also withdrawn: "obfuscation renders geometric detection and vector injection incoherent to external adversaries"; "Obfuscation defeats transfer attacks" and "A detector calibrated on uncloaked caches cannot operate on cloaked caches", true only of a fixed threshold; that the rotation result "represents transfer-attack failure (domain shift under coordinate change), not signal degradation", when it was measured with both classes under the same cloak; that this setting "is relevant only to an adversary who controls the cloak key", when it is the setting of an adversary with no key who fits a detector on cloaked caches; the heading "Adversarial Injection Is Incoherent", an untested argument presented as a result; that "KV-Cloak alone prevents external adversaries from deploying malicious geometric steering"; and that "the architecture ensures that defenders retain capabilities that attackers lose". Its title asserted a working defense, so this version has a new one. Corrected: the hedged class is 87 replies, not 89 (six ambiguous replies are excluded, and the Wilcoxon test reproduces only at 87); the spectral gap's interval is [0.810, 0.980], from the run file, not [0.806, 0.977], which no file holds; three cells of Table 2 were rounded down (0.754, 0.786 and 0.543 are 0.755, 0.787 and 0.544); the Methods said AUROCs were computed on length-residualized features, but the values reported were raw, and residualized values now stand beside them; a second analysis window, computed but never reported, is disclosed; the random-direction results were called "inconsistent with destabilization rates above 30% (power 0.97 to detect)", but 2 adverse events in 10 allow rates up to 51%, and 0.97 is the chance of seeing at least one event; the positive control's "3 calibration pairs" has no record, and the positive-control file shipped with version 1.0 has 30 pairs and no outcome labels; and the spectral gap, a raw ratio, was called Marchenko–Pastur-corrected. Version 1.0's text-only baseline (0.755, with the cache ahead in 96.7% of bootstrap samples) cannot be traced to any run file or recomputed, so this version drops it and any claim about what the cache adds to surface text. The run files behind both tables, never in this paper's folder, are added. The lab's remediation plan of 23 June 2026 recommended withdrawing the rotation claim; it was not applied. The defects were raised again by the lab's style gate (4 September 2026), Kavi's verification flags (14 September 2026), a scoping note (20 October 2026) and the paper-mode review of this correction's first draft. The withdrawn claims are the authors' responsibility. Appendix A keeps every withdrawn passage verbatim.

1 Introduction

The discovery that KV-cache geometry encodes cognitive state information creates a tension between transparency and safety. Publishing detection methods enables defensive applications (confabulation monitoring, deception detection, self-regulating AI systems) but simultaneously provides a roadmap for adversarial exploitation. An attacker with cache access could:

- **Surveil** cognitive states to identify when a model is uncertain, compliant, or vulnerable to manipulation.
- **Inject** emotion vectors calibrated to specific misalignment types, deliberately inducing confabulation, sycophancy, or overconfidence.
- **Identify** individual users or personas from cache geometry fingerprints.

This paper investigates whether KV-Cloak [Luo et al., 2026]—a lightweight, reversible cache obfuscation scheme designed for privacy protection—also mitigates these adversarial capabilities. In the forms we tested, it does not remove the signal. Under per-head rotation, the simulation used here, the feature that detects best without obfuscation still detects on both models where it was tested (Section 4.3), and an adversary can compute per-head spectra, which the rotation cannot change (Section 4.2). Under our implementation of the full mechanism, seven confabulations are too few to say whether a detectable signal survives (Section 4.7).

2 Background

2.1 The Lyra Technique

KV-cache geometry—specifically, Marchenko-Pastur-corrected spectral features derived from SVD of the cached keys and values—distinguishes cognitive states during autoregressive generation. The lab's earlier campaigns, as summarised in the Oracle Loop paper, report confabulation detection at 0.969–0.999 and hardware invariance ($r > 0.999$ across RTX 3090 and H200); that paper's own leave-one-out detection is 0.707 (0.620 after length correction) [Lyra et al., 2026c]. A within-model deception result previously reported at AUROC 1.000 across seven model scales has since been withdrawn by its authors [Lyra et al., 2026a]: a same-prompt control collapses it to 0.160, indicating the classifier separated the system-prompt template rather than deception geometry. Whether a confound-free within-model deception detector is achievable remains open; nothing in this paper depends on it. Contrastive activation addition (CAA) [Panickssery et al., 2023] enables injection of emotion vectors into the cache to steer model behavior, correcting confabulation at rates up to 96% (hostile vector on Qwen2.5-7B-Instruct, $n = 68$, McNemar $p < 0.0001$) [Lyra et al., 2026b].

2.2 KV-Cloak

Luo et al. demonstrate that an attacker can reconstruct sensitive user inputs directly from the KV cache via inversion, collision, and injection attacks. KV-Cloak mitigates this with a reversible matrix-based obfuscation scheme

combined with operator fusion, reducing reconstruction quality to random noise with minimal impact on model accuracy. The obfuscation operates inside a trusted execution environment (TEE); external consumers see only the obfuscated cache.

2.3 Threat Model

We consider an adversary who has read and/or write access to the KV cache during inference (e.g., through a shared serving infrastructure, a compromised inference endpoint, or a supply-chain attack on the serving stack). The adversary's goal is to deploy a malicious Oracle Loop: detect vulnerable cognitive states via geometric features, then inject steering vectors to destabilize the model.

The defender may optionally run a legitimate Oracle Loop inside the TEE for misalignment monitoring. The question this paper tests, whether obfuscation blinds external geometric methods, does not depend on whether a defensive Oracle is deployed.

3 Method

3.1 Obfuscation Simulation

We simulate KV-Cloak's reversible matrix obfuscation as per-head orthogonal rotation: for each attention head independently, we generate a random orthogonal matrix Q_h and apply $K'_h = K_h Q_h$, $V'_h = V_h Q_h$. This leaves the singular values of each head's key matrix unchanged, and also those of the heads concatenated along the feature axis, because the block-diagonal matrix of the Q_h is itself orthogonal. It can change the spectrum only of a matrix that combines the heads in another way. The lab's feature extractor (`spectral-shape-paper/code/lyra_features.py`) does: it stacks each layer's heads, keys and values along the token axis, so this rotation changes the spectrum it reads.

We additionally test:

- **Global orthogonal rotation**: a single Q shared across all heads per layer (preserves cross-head structure); run on the base model only.
- **Random invertible (non-orthogonal)**: $M = I + \sigma N$ where N is Gaussian.
- **Additive noise**: Gaussian noise at 1%, 5%, 10%, 50% of cache norm.
- **Diagonal scaling**: per-dimension random scaling.
- **No obfuscation**: baseline control.

- **Full KV-Cloak**: our implementation of the whole mechanism (Section 4.7).

3.2 Feature Extraction

For each trial, we extract five Marchenko-Pastur-derived features from the full KV cache (encoding + generation phases): signal rank, signal fraction, top singular value excess, spectral gap, and norm per token. The spectral gap feature (σ_1/σ_2 ratio) is not itself MP-corrected; it is a raw singular value ratio included alongside the MP-corrected features. The evaluation applies to these feature pipelines and to confabulation detection only; deception and other cognitive-state classifiers were not tested under obfuscation. AUROCs are Mann-Whitney AUROCs on the raw features; values residualized against output token count (Frisch–Waugh–Lovell, FWL; Frisch and Waugh, 1933, Lovell, 1963) are reported beside them where they matter. AUROCs are direction-free, $\max(A, 1 - A)$, so chance gives values above 0.5 at small n. Confidence intervals come from 2000 bootstrap resamples (the run files label the method "BCa percentile"); each resample is folded the same way, so every lower bound sits at or above 0.50 and the intervals do not test against chance, which the p-values do. Section 4.7 adds a label-permutation test that accounts for picking the best feature. The run files also hold a second window, the last 80 generated tokens; version 1.0 reported only the full cache, and so does this version. The script that extracted these features has not been recovered; its feature names match the lab's extractor, so we infer that it was used. The per-trial features and results are in this paper's `data/` folder (Data, Code and Review History).

3.3 Replication

A separate replication (`code/kv_cloak_replication.py`, 1,038 lines) ran 100 prompts through the distilled model: 50 asking about fabricated entities (30), unanswerable questions (10) or future events (10), and 50 factual questions. It took the cache after the prompt was encoded, before any token was generated, and at the first four full-attention layers (3, 7, 11, 15) computed six SVD features (stable rank, spectral entropy, top singular-value ratio, norm, σ_1, condition number) from the four heads' keys concatenated along the feature axis, and each head's stable rank, before and after one set of four Haar-random head rotations (seed 2026) used for every trial. A logistic-regression classifier with five-fold cross-validation (one prompt per group) was fitted to separate the two prompt types.

3.4 Models and Prompts

- **Distilled**: Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled (Jackrong). Hybrid architecture, 16 full-attention + 48 GatedDeltaNet lay-

ers. 100 confab-inducing prompts (50 about fabricated entities, 30 unanswerable, 20 about future events); behavioral classification via the Oracle Formulary classifier (40+ hedge patterns). Yields $n = 7$ confabulations and $n = 87$ hedged replies; 6 ambiguous replies are excluded.

- **Base**: Qwen2.5-7B-Instruct. Standard attention, 28 layers. Same prompt set; yields $n = 48$ confabulations, $n = 52$ hedged.

4 Results

4.1 Detection on Uncloaked Cache

On the distilled model, the spectral gap (σ_1/σ_2) achieves AUROC 0.903 (95% bootstrap CI: [0.810, 0.980]; FWL-corrected: 0.877, $R^2 = 0.068$ vs token count; $n = 7$ confabulations, $n = 87$ hedged) for confabulation detection using full-cache features. It is the best of five features and the only one that survives correction for five ($p < 0.01$); adjusted for choosing the best of the ten features in both windows, by label permutation, $p = 0.001$. Signal rank also detects (0.755 [0.578, 0.918]); the other three features are not significant ($p \geq 0.06$). The result is exploratory beyond its sample size. All seven confabulations come from fabricated-entity (4) and unanswerable (3) prompts, none from the 20 future-event prompts; six of the seven replies, and 49 of the 87 hedged ones, stopped at the 400-token cap; and in the last 80 tokens alone the spectral gap scores 0.591. The 0.903 may track prompt type or truncation as much as confabulation.

On the base model ($n = 48$ confabulations, $n = 52$ hedged), the best feature (signal fraction) achieves 0.661 (FWL: 0.687, $R^2 = 0.602$). Detection is weak, but it survives correction for five features ($p = 0.006$; after removing reply length, $p = 0.001$), on the only balanced set in this paper. The difference between models is consistent with multiple explanations including model size (27B vs 7B), architecture (hybrid vs standard), and training (Claude-distilled vs instruction-tuned). We cannot isolate the contribution of RLHF calibration from these confounds.

4.2 Rotation Leaves Per-Head Spectra Unchanged

In the replication (Section 3.3), all 24 cross-head features (six features at four layers) and all 16 per-head stable ranks were unchanged by the rotation in every one of the 100 trials, to floating-point precision (largest relative change below 10^{-6}). The classifier scored AUROC 0.994 on uncloaked features and 0.994 on cloaked features (the run file records 0.9944 for both, and a drop of 0.0), and 0.994 when fitted on uncloaked caches and tested on cloaked ones. This is an invariance check, not a detection result. The features come from

the prompt alone, the labels are prompt types (among the 50 confabulation-inducing prompts the behaviour field reads 35 confabulated and 15 hedged), and prompt length by itself separates the two prompt types at AUROC 0.968. What it does show is that these features do not move: an adversary who computes spectra per head, or across heads along the feature axis, sees the same values with or without the rotation.

4.3 The Rotation Run Behind Table 2: A Shift in Scale

In the run behind Table 2 (`data/kv_cloak_v2`), the rotation did change the features. The hedged replies' spectral gap fell from 24.7 ± 3.7 to 10.8 ± 1.3 (Wilcoxon signed-rank $W = 0$, $p = 5.5 \times 10^{-16}$, $n = 87$ paired trials); every reply moved down, while norm per token, which no orthogonal transform can change, did not move. That is what the lab's row-stacked layout predicts (Section 3.1). On the base model, a single rotation shared by all heads, which cannot change a row-stacked spectrum, left the best feature at 0.661.

The shift breaks a fixed threshold. The spectral-gap threshold that best separates the classes without the cloak (Youden's J), which flags all 7 confabulations and 21 of the 87 hedged replies, flags none of the 94 replies under the rotation, and all 94 under the full mechanism. It does not remove the signal. Within the rotation, the spectral gap still ranks confabulations above hedged replies, in the same direction, at 0.787 ($p = 0.010$; 0.765 after removing reply length, $p = 0.018$), down from 0.903. On the base model, signal fraction goes from 0.661 to 0.639 ($p = 0.017$), or from 0.687 to 0.670 after removing length ($p = 0.003$). Both are the features chosen without the cloak, so no adjustment for choosing them is needed. An adversary who refits a threshold on cloaked caches is not stopped by a change of scale, and one who computes per-head features sees no change at all (Section 4.2).

4.4 Adversarial Injection Was Not Tested

CAA [Panickssery et al., 2023] steering vectors are directions in cache space extracted via contrastive activation: $d = \mathbb{E}[\text{emotional}] - \mathbb{E}[\text{neutral}]$. Injection adds $\alpha \cdot d$ to the cache keys and values. Under per-head rotation Q_h, a vector d added to the cloaked cache becomes $d \cdot Q_h^{-1}$ in the model's own basis, a different direction if Q_h is unknown to the attacker. That is an argument, not a result. The test of it was uninterpretable: direct steering, the positive control, corrected 0–1 of 10 replies on each of four models, and on Mistral a random direction of the same norm gave exactly the same counts as the steering vector (1 corrected, 2 adverse, 7 unchanged). The files record only these counts, not which prompts the ten trials used or how many could have been corrected. An injection that does nothing without the cloak cannot show what the cloak does to it.

4.5 Within-Cloak Detection

With both classes under the same rotation, detection on the distilled model is 0.787 (Table 2). This is the setting of an adversary who has no key but can label some cloaked caches, for example from the model's outputs, and fit a detector on them; an adversary who holds the key can remove the cloak and use the uncloaked 0.903. Across the five models with a detection result (Table 1), within-rotation detection was lower on four (by 0.02 to 0.19; no test of these differences was run) and higher on one: on Qwen2.5-14B it went from 0.784 to 0.951 ($p < 0.001$). We have no explanation for the rise. On Mistral it fell from 0.707 to 0.516, but the uncloaked 0.707 does not survive correction for four comparisons. Under each of the other transforms run on the distilled model (random invertible, diagonal scaling, Gaussian noise at four levels), the best of the five features stayed at 0.765 or above, which is about what choosing the best of five gives by chance at this size; under diagonal scaling the spectral gap itself fell to 0.517.

4.6 Cross-Architecture Results

Detection and obfuscation effects vary across architectures (Table 1).

Table 1: Cross-architecture KV-Cloak results. Prompts per model: 100 for Qwen3.5-27B-Dist. ($n = 7$ confabulations) and Qwen2.5-7B, 50 for the other four. Det.: the best single feature (spectral gap and signal fraction for the two Qwen models; the cross-architecture files do not name it). Det.(clk) is detection with both classes under the per-head rotation.

Model	Confab%	Det.	Det.(clk)	Mag. adv.
Qwen3.5-27B-Dist.	7%	0.903	0.787	—
Qwen2.5-14B	12%	0.784	0.951*	1/10
Mistral-7B	26%	0.707	0.516	2/10
Llama-3.1-8B	14%	0.590	0.559	1/10
Qwen2.5-7B	48%	0.661	0.639	—
DeepSeek-R1-7B	2%	—	—	0/10

*Rises under the rotation; unexplained.

Mag. adv. = adverse events from magnitude attack (random direction, 10 trials).

Uncloaked detection ranges from 0.590 (Llama) to 0.903 (distilled Qwen); no cross-architecture model survives correction for four comparisons. Perturbation along a random direction, at the norm of a steering vector, produced 0–2 adverse events per model in 10 trials, 4 of 40 over the four models. Ten trials exclude little: 2 adverse events in 10 are compatible with per-trial rates up to 51%, and the files do not record which trials could have had one.

4.7 Simulation vs Real KV-Cloak Implementation

We implemented the full KV-Cloak mechanism ($K' = S \cdot \hat{P} \cdot (K + A) \cdot M$) following Luo et al. and compared it against our per-head orthogonal rotation simulation on the same 100 prompts (Table 2). Our implementation follows their description; it is not the authors' code.

Table 2: Detection AUROC (raw features, not residualized for reply length): uncloaked vs simulation vs real KV-Cloak (distilled model, $n = 7$ confab, $n = 87$ hedged). Bootstrap 95% CIs.

Feature	Uncloaked	Sim. rotation	Real KV-Cloak
spectral_gap	**0.903** [.81,.98]	0.787 [.58,.95]	**0.614** [.51,.81]
signal_rank	0.755 [.58,.92]	0.770 [.62,.90]	0.766 [.57,.94]
signal_fraction	0.714 [.52,.90]	0.683 [.51,.88]	0.681 [.51,.88]
top_sv_excess	0.609 [.51,.79]	0.544 [.50,.76]	0.736* [.53,.92]
norm_per_token	0.598 [.51,.77]	0.598 [.51,.77]	0.673 [.52,.85]

*Rises under the mechanism; see text.

The full mechanism moves σ_1: the additive mask raises the spectral gap of every reply, from a hedged mean of 24.7 to 84.6 or more. The spectral gap, the best feature without the cloak, falls to 0.614 ($p = 0.33$; 0.593 after removing reply length). Signal rank (0.766, $p = 0.020$) and top singular-value excess (0.736, $p = 0.038$) are no lower than without the cloak (0.755 and 0.609). But signal rank mostly tracks reply length ($R^2 = 0.77$ against output token count, with or without the cloak), and any cloak preserves length, because the cache has one row per token. After removing length, the best feature under the mechanism is signal rank at 0.705 (top singular-value excess 0.695), and adjusted for choosing the best of five, $p = 0.19$; on raw features the best is 0.766, with $p = 0.054$. With seven confabulations, the best of five raw features reaches 0.77 in 5% of label shuffles. The mechanism removes the one feature that detected well. Whether a confabulation signal survives it, these data cannot say.

The two σ_1-dependent features. Both changes come from the additive mask raising σ_1: it lowers the spectral gap (0.903 to 0.614) and raises top singular-value excess (0.609 to 0.736). With seven confabulations we cannot tell whether 0.736 is class information surviving the mask or noise (unadjusted $p = 0.038$; 0.695 after removing reply length).

5 Discussion

5.1 What This Means for Dual Use

Version 1.0 argued that KV-Cloak mitigates the Lyra Technique's dual-use risk, so that cache-geometry methods could be published while deployed models stayed protected. The results do not support that. A rotation-based cloak leaves per-head spectra unchanged, and in the lab's own layout it shifts the scale of the features but leaves the confabulation ranking in place on both models where it was tested; it stops only a detector whose threshold is never refitted. Our implementation of the full mechanism removed the best feature, but with seven confabulations we cannot tell whether it removed the signal. Nothing shown here mitigates the dual-use risk of publishing cache-geometry methods. A defense would have to be shown on its own terms: against adaptive adversaries who choose their features after seeing the cloak, with enough confabulations to measure what remains.

5.2 A Defender Inside the TEE (an Argument)

A defender operating inside the TEE could run the Oracle Loop before obfuscation, so its detection would be the uncloaked detection (AUROC 0.903 on the distilled model) and its steering vectors would be injected in the model's own basis. That is an argument, not a tested system, and it has a gap: in KV-Cloak the feature matrix M is fused into the model weights, so the cache the defender sees is already transformed by M, and the Oracle Loop paper notes that detection under fused M has not been verified [Lyra et al., 2026c]. It gives defenders an advantage only if attackers lose the signal, which the results above do not show.

5.3 What a Test of the Architecture Would Need

The deployment architecture version 1.0 recommended remains a proposal, untested here:

1. KV-Cloak obfuscation active on all external cache access.
2. Oracle Loop (if deployed) operating inside the TEE on the pre-obfuscation cache.
3. Obfuscation key material and Oracle parameters both TEE-protected.
4. Per-session or per-query key rotation to prevent statistical recovery from accumulated observations.

The replication used one rotation for every trial, so per-query key rotation (item 4) was not exercised there, and the run behind Table 2 does not record whether its key changed per trial. Nothing here measured what it costs an adversary to recover the signal.

5.4 Limitations

1. **Small sample size.** The distilled model yields only $n = 7$ confabulations from 100 prompts. The headline AUROC 0.903 has bootstrap CI [0.810, 0.980] and is the best of five features; adjusted for that choice it remains significant, but it may track prompt type or truncation (Section 4.1), and the comparisons under the full mechanism (Section 4.7) are not powered. Confirmation on a higher-confab-rate model is needed.

2. **Feature code not recovered.** The script behind Tables 1 and 2 has not been recovered. Its per-trial outputs are in `data/`; that it used the lab's row-stacked extractor is inferred from its feature names and from how the features respond to the two rotations.

3. **Cache versus text untested.** Version 1.0's text-only baseline cannot be traced or recomputed (Correction note), so nothing here shows what the cache adds to surface text features.

4. **TEE trust assumption.** The defensive architecture assumes TEE integrity. Documented side-channel attacks against SGX (Van Bulck et al., 2018; Schwarz et al., 2019) and SEV (Li et al., 2021) mean this is a hardware trust assumption, not a cryptographic guarantee. The security reduction is to TEE integrity, not to a standard hardness assumption.

5. **Injection test underpowered.** Direct steering, the positive control, corrected only 0–1 of 10 replies per model, too few to establish a working baseline. Through-cloak injection results are therefore uninterpretable: we cannot distinguish "cloak defeats injection" from "injection was never effective." The 30-pair positive-control run shipped with this paper (`data/kv_cloak_positive_control.json`) stored replies without outcome labels and is not analysed here.

6. **Cross-architecture significance.** No individual model in the cross-architecture sweep achieves significance after Bonferroni correction for four comparisons. Detection is individually suggestive on Qwen-14B (0.784) and Mistral (0.707) but should not be interpreted as confirmed cross-architecture replication.

7. **Adaptive adversary untested.** All tests assume a static, non-learning attacker. An adversary who can submit probe queries and observe behavioral outputs could iteratively estimate the rotation basis. Formal query-complexity bounds are not provided. The threat model is bounded to single-query, non-adaptive adversaries.

6 Conclusion

Per-head rotation, the simulation of KV-Cloak this paper relied on, does not remove the confabulation signal: on both models where it was tested, the feature that detected best without it still detects with it, and an adversary who computes spectra per head sees no change at all. Our implementation of the full mechanism removed the best detection feature on the distilled model, and after removing reply length no other feature detects significantly; with seven confabulations we cannot say whether a detectable signal remains. Version 1.0 concluded that KV-Cloak and the Oracle Loop are complementary, with defenders keeping what attackers lose; nothing here shows that attackers lose the signal. Whether any reversible obfuscation blinds cache-geometry detection is open, and a test of it needs adaptive adversaries, per-query keys and many more confabulations.

Data, Code and Review History

Tables 1 and 2: `data/kv_cloak_v2/`, `data/kv_cloak_real/`, `data/kv_cloak_cross_arch/` and `data/kv_cloak_base/`, added in version 1.1, copied unchanged from the lab's KV-Experiments repository (`results/oracle_harness/`, commit 8429260, May 2026; that repository is private). The scripts that wrote them were not recovered. Replication: `code/kv_cloak_replication.py` with `data/kv_cloak_replication_100.json` and `data/kv_cloak_replication_analysis.json`. Positive control: `data/kv_cloak_positive_control.json`. Every result in this version is recomputed from the per-trial files, or read from the run files where only aggregates exist (Table 1's cross-architecture rows, all confidence intervals), by `check_correction_claims.py`, in this paper's folder of `https://github.com/Liberation-Labs-THCoalition/published-research`; Methods details carried over from version 1.0 (model architectures, the classifier's pattern count) are not. Review history: the lab's remediation plan (23 June 2026) recommended withdrawing the rotation claim; the Agni style gate (4 September 2026) rejected version 1.0 (`tools/agni/style/kv-cloak-defense-paper.agni_style.json`); Kavi's verification flags (14 September 2026); for version 1.1, a scoping note (published-research pull request 10, 20 October 2026), the correction gate above, and the Agni paper and style modes on the first draft (records in `review/`).

Acknowledgments

We thank Dwayne Wilkes (Sentient Futures / Liberation Labs) for statistical auditing and red-team review.

References

Ragnar Frisch and Frederick V Waugh. Partial time regressions as compared with individual trends. *Econometrica*, 1(4):387–401, 1933. doi: 10.2307/1907330.

Michael C Lovell. Seasonal adjustment of economic time series and multiple regression analysis. *Journal of the American Statistical Association*, 58 (304):993–1010, 1963. doi: 10.1080/01621459.1963.10480682.

Zhifan Luo, Shuo Shao, Su Zhang, Lijing Zhou, Yuke Hu, Chenxu Zhao, Zhihao Liu, and Zhan Qin. Shadow in the cache: Unveiling and mitigating privacy risks of KV-cache in LLM inference. In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2026. doi: 10.14722/ndss.2026.240258. arXiv:2508.09442.

Lyra, Thomas Edrington, and CC. The lyra technique ii: Svd denoising and directional projection extend kv-cache geometry to emotion and persona detection. 2026a. doi: 10.5281/zenodo.20604329. Liberation Labs / THCoalition.

Lyra, Thomas Edrington, and Dwayne Wilkes. The oracle formulary: Emotion-vector dose–response profiles for kv-cache steering. *Liberation Labs Technical Report*, 2026b. URL `https://liberationlabs.tech`.

Lyra, Thomas Edrington, and Dwayne Wilkes. The oracle loop: Cache-geometry detection and steering of confabulation in language models. *Liberation Labs Technical Report*, 2026c. URL `https://liberationlabs.tech`.

Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2023. doi: 10.48550/arXiv.2312.06681.

A Correction Record (October 2026)

This appendix keeps the withdrawn wording of version 1.0 visible. Each item quotes it verbatim and says where the evidence is.

1. **Title.** "Cache Geometry Under Obfuscation: KV-Cloak as Defense Against Adversarial KV-Cache Steering." It asserted a working defense (Sections 4.3 and 4.7).

2. **Abstract, opening.** "A single invertible linear transform of the KV cache degrades SVD-based cognitive-state detection from near-ceiling to near-chance ($p = 5.5 \times 10^{-16}$), while a real KV-Cloak implementation confirms the defense is conservative (detection AUROC drops from 0.903 to 0.614)." The p-value is the Wilcoxon test of Section 4.3, a shift in one feature's values; under the rotation the same feature still detects at 0.787 (Section 4.3); under the full mechanism, seven confabulations cannot settle whether the signal is gone (Section 4.7).

3. **Abstract, finding 1 (text baseline).** "A text-only surface-feature baseline achieves 0.755; cache geometry outperforms text in 96.7% of bootstrap samples (delta 0.148, 95% CI [−0.008, 0.322])—suggestive but not significant at the 95% level." Results: "The best text-only feature (average word length) achieves 0.755 [0.560, 0.924]. Cache geometry outperforms text in 96.7% of bootstrap samples, but the 95% CI on the difference includes zero (delta: 0.148 [−0.008, 0.322])." No run file holds these numbers, and the stored trials keep only the first 150 characters of each reply.

4. **Abstract, finding 2.** "**Obfuscation defeats transfer attacks.** [. . .] A detector calibrated on uncloaked caches cannot operate on cloaked caches. This is a domain-shift effect: rotation preserves the underlying geometric structure but changes the coordinate system, preventing transfer from uncloaked calibration data." True only of a fixed threshold; the ranking survives the rotation, and in a per-head layout nothing moves (Sections 4.2 and 4.3).

5. **Abstract, finding 3.** "0–2 adverse events per model in 10 trials, inconsistent with destabilization rates above 30% (power 0.97 to detect) but unable to rule out rates up to 20% at this sample size" and "(3 calibration pairs; direct injection produced 0–1/10 corrections)". Two events in 10 allow rates up to 51%; 0.97 is $1 - 0.7^{10}$, the chance of at least one event; no record gives 3 calibration pairs.

6. **Abstract, closing.** "We conclude that KV-Cloak and the Oracle Loop are complementary: the Oracle Loop provides misalignment detection and correction; KV-Cloak prevents external adversaries from weaponizing the same geometric methods. Together, they address the dual-use risk inherent in publishing cache-geometry techniques." And: "The rotation-simulation result (0.786) represents transfer-attack failure (domain shift under coordinate change), not signal degradation."

The 0.786 (0.787) was measured with both classes under the same rotation, not by transferring a detector.

7. **Introduction.** "We show that it does: obfuscation renders geometric detection and vector injection incoherent to external adversaries, while a defender operating inside the trusted execution environment retains full access to the uncloaked cache and its geometric properties."

8. **Threat Model.** "[. . .] but our primary result—that KV-Cloak blinds external geometric methods—holds regardless of whether a defensive Oracle is deployed."

9. **Method.** "This preserves within-head singular values while scrambling cross-head correlations." Concatenated along the feature axis, the heads' singular values are preserved too; stacked along the token axis, as the lab's extractor stacks them, they are not (Section 3.1). Also: "Features are FWL-residualized [. . .] against output token count and evaluated via Mann-Whitney AUROC"; the AUROCs reported were computed on raw features.

10. **Results, feature-space shift.** "The cloaking is perfectly detectable by paired comparison—but this is the desired property. An external adversary's detection model, calibrated on uncloaked caches, encounters a completely alien feature distribution. Any detection thresholds, classifier boundaries, or anomaly scores calibrated on uncloaked data are meaningless on cloaked data." (Section 4.3.)

11. **Results, injection.** The heading "Adversarial Injection Is Incoherent" and "The injection becomes noise rather than a targeted perturbation." Untested: the positive control did not work.

12. **Results, defender.** The heading "Defensive Oracle Retains Full Capability" and "Detection operates at full capability (AUROC 0.903 on the distilled model)." An argument, now in the Discussion with the fused-M gap.

13. **Results, within-cloak detection.** "On Mistral, the cloak effectively eliminates detection entirely. [. . .] This scenario—where both classes share the same obfuscation—is relevant only to an adversary who controls the cloak key, which the TEE architecture prevents."

14. **Results, base model.** "Detection is weak and heavily confounded with token count." Removing reply length raises it (0.661 to 0.687).

15. **Results, cross-architecture.** "Obfuscation reduces detection on Mistral to chance (0.516) and partially degrades on the distilled

model (0.786 under simulation). Magnitude-based attacks (direction-agnostic perturbation) produce at most 2 adverse events in 10 trials across all architectures—models are robust to random perturbation at steering-vector magnitude."

16. **Results, simulation vs implementation.** "Both degrade a transfer attack—a detector calibrated on uncloaked caches applied to cloaked caches—rather than destroying the underlying geometric signal." On top singular-value excess: "This is an artifact of the mask magnitude, not a genuine detection signal, and would not generalize to unseen mask instances under per-query key rotation"; the table note "Anomalous increase; likely artifact of additive mask A inflating top singular values differentially across behavioral classes"; and "the defense works *by* perturbing σ_1". Per-query rotation was not tested.

17. **Discussion.** "The Lyra Technique's dual-use risk—that published cache-geometry methods could enable adversarial steering—is mitigated by KV-Cloak. Cache obfuscation renders both geometric detection and vector injection incoherent to external adversaries. A model operator deploying KV-Cloak can publish Oracle Loop methods transparently while protecting deployed models from exploitation via those same methods." "This addresses the responsible-disclosure tension identified by Watson (2026): the defense can be published alongside the technique, enabling the research community to build on the methods while preventing weaponization against production systems." "Without a defensive Oracle Loop, KV-Cloak alone prevents external adversaries from deploying malicious geometric steering."

18. **Conclusion.** "KV-Cloak and the Oracle Loop are complementary technologies. The Oracle Loop enables detection and correction of misalignment through KV-cache geometry. KV-Cloak prevents external adversaries from exploiting those same geometric methods. Together, they address the dual-use risk inherent in publishing cache-geometry research: the technique is transparent, the defense is available, and the architecture ensures that defenders retain capabilities that attackers lose."

19. **Academic editions only.** The abstract of the academic file `academic_main.tex`: "degrades the strongest detection feature (spectral gap) from 0.903 to 0.614—near chance—confirming that the defense is stronger than our initial orthogonal-rotation simulation (0.786) suggested." Its Results: "confirming that per-head orthogonal rotation alone *understates* the defense. [...] The simulation was conservative: the actual defense is stronger than we initially reported." A limitation in both academic files: "The real mechanism degrades spectral

gap detection more than the simulation (0.614 vs 0.786), confirming that the simulation was conservative." Another, in both: "Until resolved, the scope of the obfuscation-degrades-detection claim excludes this model" (Qwen2.5-14B); the claim it scoped is withdrawn, and the rise is reported in Section 4.5.